

PICASSO 2025: 8th International Workshop on Physics Embedded AI Solutions in Mobile Computing

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Abstract

Real-world ubiquitous computing systems often encounter challenges in acquiring sufficient, high-quality data for accurate inference using purely data-driven methods. The efficacy of such methods is highly dependent on both the quantity and quality of available data. However, practical constraints frequently limit data collection, making it costly or even infeasible. Incorporating physical knowledge—comprising domain expertise, experiential heuristics, and analytical models—offers a promising avenue to mitigate these limitations. This workshop aims to foster exploration at the intersection of data-driven methods and physical knowledge. It will unite domain experts who understand the underlying physical phenomena, system developers, and researchers from traditional data-driven disciplines. We invite submissions addressing these challenges across diverse applications and domains, as well as algorithmic and systematic approaches leveraging physical knowledge. Our goal is to cultivate a community dedicated to systematically analyzing data quality for inference purposes and evaluating the enhancements achievable through physical knowledge. Preliminary and ongoing research contributions are welcome.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

Keywords

ubiquitous computing; data-driven; physical knowledge; cyber-physical system

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1 Objective of the Workshop

The workshop aims to systematically discuss and advance methods integrating physical knowledge into purely data-driven ubiquitous computing systems [3], specifically:

- Enabling accurate learning with limited labeled data [4].
- Investigating the impact of physical knowledge on computing performance [2].
- Exploring applications of Large Language Models (LLMs) and Embodied AI in ubiquitous computing contexts [1].

These objectives directly address inherent challenges faced by purely data-driven methods, such as:

- Limited labeled data availability in real-world scenarios.
- Spatiotemporal variability of data estimation accuracy.

Therefore, more and more researchers are looking into physical knowledge, including **heuristics, domain knowledge, and analytical models**, to achieve a better understanding of the data and eventually a higher learning accuracy. However, existing work that falls into this category still lacks a systematic solution. This emerging area needs a systematic approach to enhance the now-widely-used pure data-driven approaches.

We plan to achieve this goal by bringing together researchers who are faced with these types of problem in different domains or working on various applications. The discussion on the limitation of the data and how to utilize the physical knowledge to enhance the ubiquitous computing performance. Various applications may face this type of data limitation in different ways. Therefore, bringing problems from different perspectives together allows for a broader discussion on this issue and may enable the development and generalization of these individual issues.

2 Relevant and Timely

Ubiquitous computing research prioritizes human-centric, multi-modal data from heterogeneous sensing systems spanning city-wide to individual levels. As this field expands, the limitations of purely data-driven approaches have become apparent, often due to the insufficient coverage of diverse real-world conditions. Consequently, interdisciplinary approaches that integrate physical and data-driven knowledge have emerged to improve inference performance.



Figure 1: Group photo of PICASSO 2024 participants.

The UbiComp community, known for its multidisciplinary emphasis on design, development, and deployment, provides an ideal platform for exploring these intersections. The multidisciplinary spirit of the conference fits the topic of our workshop very well. Many applications that target understanding the human and the environment, a.k.a. the physical world, face the problem of obtaining sufficient (label) data to infer the status/information accurately.

There is no systematically approach that generalizes this combination of physical and data-driven knowledge because of the complexity of individual problems. In addition, the forms of these problems vary significantly for different applications or domains. With the growth of system studies and machine learning studies, we hope this workshop can start a new subfield of computing that combines physical knowledge and data-driven approaches to achieve a balanced system performance for various applications.

Previous workshops. Building on successful CPD workshops at UbiComp 2018-2023 and PICASSO workshop at MobiCom 2024, our series has consistently highlighted limitations in purely data-driven methods. In the past four years, we received 100+ submissions from Asia, Australia, Europe, and North America, which show the worldwide impact of the workshop. In addition, we attracted more than 50 audience to attend our workshop last year (shown in Figures 1 to 3). Based on the historical numbers of both submissions and attendees, we expect a growing trending and higher popularity this year.

Highlighted topics from previous workshops include:

- System sensibility
- Understand and fusing data
- Using physical knowledge to enhance neural network
- Human centric learning estimation with pure data-driven approaches.

Each drew a lot of attention and participants had heated discussions over these topics.

3 Organizers and Program Committee

3.1 Organizers

- **Julio Sahuquillo Borrás** is a Professor in the Department of Computer Engineering at the Universitat Politècnica de València, Spain. He holds a Ph.D. in Computer Engineering from the same institution. His research focuses on computer architecture, with an emphasis on processor design, memory hierarchy, and performance evaluation. He has contributed to the development of NOREBA, a processor architecture for



Figure 2: Keynote Speaker of PICASSO 2024: Jian Song, IEEE Fellow, Tsinghua University



Figure 3: PICASSO 2024 Oral Presentation Session

non-speculative out-of-order retirement via branch reconvergence analysis.

- **Fan Dang** is an Assistant Professor at the School of Software Engineering at Beijing Jiaotong University. He holds a Ph.D. in Software Engineering from Tsinghua University, where he also completed his Bachelor's degree in Computer Software. His research focuses on Industrial Intelligence and Edge Computing. Dr. Dang serves as an Associate Editor for the ACM Transactions on Sensor Networks (TOSN) and Tsinghua Science & Technology.
- **Miloš Stojmenović** is a Professor of Computer Science at Singidunum University in Belgrade, Serbia. He earned his Bachelor's degree in Computer Science from the University of Ottawa (1999-2003), pursued postgraduate studies at

Carleton University (2003-2005), and completed his doctoral dissertation at the University of Ottawa (2005-2008). His research interests include Computer Vision, Image Processing, Cloud Computing, and Wireless Networks.

- **Pengfei Zhou** is an Assistant Professor in the Department of Informatics and Networked Systems at the University of Pittsburgh's School of Computing and Information. He leads the Mobile Intelligence and Networking Technology (MINT) research group, focusing on mobile sensing and networking, Artificial Intelligence of Things (AIoT), and autonomous cyber-physical systems.
- **Jingao Xu** is a Postdoctoral Researcher at Carnegie Mellon University's Computer Science Department, collaborating with Professor Mahadev Satyanarayanan in the Living Edge Lab. His research focuses on advancing edge computing within drone-based systems. He earned both his Ph.D. (2022) and B.E. (2017) degrees from Tsinghua University. Dr. Xu's research interests include Artificial Intelligence of Things, edge computing, and drone-based computing. Dr. Xu has been recognized for his contributions to the field, serving as the Finance Co-Chair for ACM MobiSys 2025.

3.2 Program Committee

- Susu Xu (Johns Hopkins University)
- Lingkun Li (Beijing Jiaotong University)
- Zhengxiong Li (University of Colorado Denver)
- Linqi Song (City University of Hong Kong)
- Yongpan Zou (Shenzhen University)
- Weitao Xu (City University of Hong Kong)
- Shijia Pan (University of California Merced)
- Mostafa Mirshekari (Stanford University)
- Shahab Bahrami (The University of British Columbia)
- Marzieh Khakifirooz (Tecnológico de Monterrey, Mexico)
- M. Hadi Amini (Florida International University)
- Roozbeh Jafari (Texas A&M University)
- Pan Hu (Stanford University)
- Miadreza Shafie-khah (University of Vaasa)
- Sicong Liu (Northwestern Polytechnical University)
- Wenbo Ding (Tsinghua University)
- Linqi Song (City University of Hong Kong)
- Le Yang (Xi'an Jiaotong University)
- Xinlei Chen (Tsinghua University)
- Yi Ding (The University of Texas at Dallas)
- Yu Yang (Lehigh University)
- Chunhui Duan (Beijing Institute of Technology)
- Dongyue Huang (The Chinese University of Hong Kong)
- Pengjin Xie (Beijing University of Posts and Telecommunications)
- Jinliang Yuan (Tsinghua University)

3.3 Steering Committee

The Steering Committee members are experienced conference organizers who can provide feedback and guidance to workshop organization.

- Yunhao Liu (Tsinghua University, ACM Fellow)
- Pei Zhang (University of Michigan)

4 Workshop Papers

4.1 Important Dates

- Submission Deadline: June 15, 2025
- Author Notifications: June 29, 2025
- Camera-ready Deadline: July 30, 2025

4.2 Planned Review Procedure

Each paper will be reviewed by at least three members. The criteria for accepted papers are consistent with our first workshop, including

- if the paper address or reveal the “limited data” problem for pure data-driven approaches
- if the physical knowledge is used to enable better information inference
- if the paper/solution/observation is controversy or able to trigger discussion in depth regarding the combination of physical and data-driven knowledge

4.3 Call For Papers

The goal of the workshop is to explore the intersection between (and the combination of) data and physical knowledge. The workshop aims to bring together domain experts that explore the physical understanding of the data, practitioners that develop systems and the researchers in traditional data-driven domains. The workshop welcomes papers which focus on addressing these issues in different applications/domains as well as algorithmic and systematic approaches to applying physical knowledge. Therefore, we further seek to develop a community that systematically analyzes the data quality regarding inference and evaluates the improvements from physical knowledge. Preliminary and on-going work is welcomed.

Topics of interests include, but are not limited to:

- Innovations in learning algorithms that combine physical knowledge or models for sensor perception and understanding
- Experiences, challenges, analysis, and comparisons of sensor data in terms of its physical properties
- Sensor data processing to improve learning accuracy
- Machine learning and deep learning with physical knowledge on sensor data
- Mobile and pervasive systems that utilize physical knowledge to enhance data acquisition
- System services such as time and location estimation enhanced by additional physical knowledge
- Heterogeneous collaborative sensing based on physical rules
- Distributed sensing for cyber-physical systems
- Advanced machine learning algorithms and solutions for efficient sensing

The application areas include but not limited to:

- Human-centric sensing applications
- Environmental and structural monitoring
- Smart cities and urban health
- Health, wellness & medical
- Smart energy systems and intelligent transportation networks
- Large Language Model and its applications in mobile system

Authors must cite and relate their submissions to relevant prior publications of their own. If applicable, ethical approval for experiments with human subjects should be demonstrated as part of the submission.

5 Target Attendees and Expected Numbers

We seek original work on:

- addressing specific problems arising from data limitations through leveraging physical knowledge of those limitations,
- observations of physical phenomena that cause data limitations or inference challenges across multiple applications,
- introducing emerging data-driven techniques at the intersection of ubiquitous computing and machine learning.

The first direction allows the audience to deepen their understanding of how physical knowledge can inform inference from limited data, while the second encourages the exploration of more generalized approaches. Since data limitations often occur in sensing data from physical systems, our target authors are primarily (though not exclusively) from the UbiComp, CPS-IoT Week, and SenSys/BuildSys communities.

We expect 20 to 30 submissions, with an acceptance target of 10 to 18 papers. The program will include two sessions, each featuring four to six presentations of accepted papers. A 15-minute break

will be scheduled between consecutive sessions to allow timely discussions between authors and attendees.

6 Planned Publicity Procedure

We will distribute the CFP through relevant email lists as well as direct emails to our research colleagues. The CFP will be circulated on the following dates: 04/30/2025, 05/15/2025, and 06/01/2025. In addition to email distribution, we will also promote the workshop through social media platforms, including Facebook, X (Twitter), LinkedIn, and WeChat.

7 Planned Workshop Activity

For each accepted original short paper, there will be a 12 min presentation followed by 3 mins Q & A or discussion.

References

- [1] 2022. On the Opportunities and Risks of Foundation Models. arXiv:2108.07258 [cs.LG] <https://arxiv.org/abs/2108.07258>
- [2] George Em Karniadakis, Ioannis G Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. 2021. Physics-informed machine learning. *Nature Reviews Physics* 3, 6 (2021), 422–440.
- [3] Jared Willard, Xiaowei Jia, Shaoming Xu, Michael Steinbach, and Vipin Kumar. 2022. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Comput. Surv.* 55, 4, Article 66 (Nov. 2022), 37 pages. <https://doi.org/10.1145/3514228>
- [4] Zhi-Hua Zhou. 2017. A brief introduction to weakly supervised learning. *National Science Review* 5, 1 (08 2017), 44–53. <https://doi.org/10.1093/nsr/nwx106> arXiv:<https://academic.oup.com/nsr/article-pdf/5/1/44/31567770/nwx106.pdf>